LDCrowdNet: A Lightweight Network for Dense Crowd Counting

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Abstract. Counting crowds is essential for applications such as urban planning, traffic monitoring, and video surveillance. However, accurately counting the number of individuals (people) is challenging within an image, especially for real-time applications. Recently developed Dense Crowd Counting systems incur high computational costs due to the utilize of computation intensive deep networks. Accordingly, this work proposes a novel lightweight model named Lightweight Network for Dense Crowd (LDCrowdNet), specifically designed for crowd counting and crowd density estimation. LDCrowdNet integrates components such as a ShuffleNetV2 backbone network, a proposed novel enhanced ECANet for efficient channel attention, an introduced feature pyramid network (FPN), and a context module (CM) that serves as a Density Map Generator (DMG). Extensive experiments have been conducted across multiple benchmark datasets, illustrate that LDCrowdNet surpasses current state-of-the-art methods in terms of accuracy and robustness. This superiority is evident across multiple crowd counting datasets: ShanghaiTech, UCF_CC_50, and UCF-QNRF.

Keywords: Crowd Counting · Density Estimation · Convolutional Neural Networks (CNN) · Lightweight Network.

1 Introduction

Counting the number of individuals (people) is crucial in congested crowded scenes for applications like public protection [35], traffic monitoring [11], urban planning [12], and business promotion [30]. Crowd counting involves estimating the count of individuals within an image in computer vision tasks. This task has grown more significant due to the challenges including occlusion, scale variation, and background clutter. Although earlier approaches relied on detection-based and regression-based techniques, CNN-based density map estimation approaches have become mainstream for their robust feature representation.

A range of network models [32,12,34,1,23] has been developed for crowd flow monitoring, crowd control, and security enhancement. The analysis of congested scenes has advanced ranging from basic crowd counting to density map representation, which illustrates the characteristics of the crowd distribution [33] providing more precise information crucial for high-risk environments. The evolution of methods for analyzing congested scenes has transitioned from straightforward crowd counting, which provides the total count of individuals within an image to density map representation (which illustrates the traits of the crowd dispersion [33]). CNN-based approaches have become mainstream due to their robust feature representation capabilities.

Counting objects in images has led to various techniques in CNN-based methods [16], [15]. Traditionally, researchers have favoured heavyweight networks like VGGNet [22], ResNet [6], and HRNet [25] as backbones for extracting the subtle features. Developing lightweight models entails crafting concise network architectures to diminish computational complexity, parameters, and execution time. Techniques such as low-rank pruning, knowledge distillation, and model compression facilitate the creation of lightweight neural networks for crowd counting tasks, exemplified by models like MobileNets [7], MobileNetV2 [21], and Mobile-Count [27].

Despite their achievements, these models still face common issues. Popular lightweight networks sacrifice convolutional layers to achieve faster inference speeds, which can limit their ability to effectively extract crowd features. This reliance on a unique prominent feature for density maps may lead to a loss of pixel information for small objects. Many lightweight networks use feature maps from several stages without entirely integrating them, resulting in a loss of semantic contextual information. This limitation raises a significant hurdle for researchers.

To tackle these hurdles, we introduce LDCrowdNet, a lightweight network for dense crowd counting, using ShuffleNetV2 [19] as the backbone to capture crowd features across scales. Our focus is on improving the Efficient Channel Attention module in CNNs for enhancing the efficiency and performance, prioritizing relevant feature extraction without compromising efficiency.

Our key contributions to this work can be as follows:

- We introduce a lightweight dense crowd counting method, LDCrowdNet, which reduces model parameters and complexity. ShuffleNetV2 [19] is used as the backbone network for simplifying the architecture.
- In the front end of our network, we present an enhanced Efficient Channel Attention (ECA) module, ECANet, to prioritize relevant feature extraction from backbone crowd counting network channels, avoiding dimensionality reduction. This module is integrated into architectures like Feature Pyramid Network (FPN) and Context Module (CM) to generate higher-quality density maps for density estimation.
- In the back end of our network, utilize the Feature Pyramid Network (FPN) to fuse multi-stage features extracted from the channel attention network. The Context Module (CM) acts as a crowd detector, identifying annotated heads at various scales in a single pass. Additionally, the CM functions as a Density Map Generator (DMG) that generates higher-quality density maps to facilitate density estimation.

In summary, our approach significantly improves the ECA module, resulting in refined accuracy and efficiency. LDCrowdNet reduces the parameters and complexity of the model while boosting the execution for dense crowd counting tasks.

The remaining parts of this paper is summarized as follows: Section section 2 explores the related work on crowd counting. Section section 3 delivers an elaborate explanation of the proposed method. Section section 4 outlines the implementation and training details, and Section section 5 demonstrates the experimental results, including an analysis of the proposed method and an ablation study. The paper concludes in Section section 6.

2 Related Work

Early crowd counting approaches predominantly depended on detection [26] and regression [8] techniques. However, these methods encountered challenges in crowded scenes, frequently falling short of achieving sufficient accuracy due to inadequate feature extraction. Recently, methods based on density map estimation within deep learning frameworks have achieved high levels of accuracy. However, many of these approaches utilize computation intensive deep networks. This work focuses on the development of lightweight networks (with floating point operations lesser than 20 GFLOPs) for crowd density estimations. Accordingly, only related works are briefly reviewed next. The broader literature in crowd counting is vast. Interested researchers may refer to the review papers on crowd counting [3,10,24].

Recent works have focused on designing lightweight networks for crowd counting tasks. Models like MobileNets [7] are designed for mobile and edge device applications. These lightweight networks utilize depthwise separable convolutions technique for network development. The MobileNetV2 [21], an updated version, includes lightweight depthwise convolutions in intermediate dilation layers for feature filtering. Another lightweight model [18] includes a simple an encoder and decoder of image feature to convey density information effectively without adding more convolutional layers. MobileCount [27], based on MobileNetV2, enhances the counting performance without increasing computational complexity (floating-point operations). ShuffleNetV2 [19], optimized for mobile devices, offers high efficiency and accuracy. Inspired by these methods, we employed a ShuffleNetV2 as the lightweight backbone for our proposed LDCrowdNet due to its impressive performance.

3 Proposed Method

The proposed LDCrowdNet architecture, as illustrated in Fig. 1, consists of four primary components: 1) a backbone network using ShuffleNetV2 [19] as the feature extractor; 2) an enhanced ECANet, which is an efficient channel attention network; 3) feature pyramid network (FPN), and 4) a context module (CM). Backbone ShuffleNetV2 [19] captures the input crowd image $I \in \mathbb{R}^{H \times W \times 3}$ of resolution $R : H \times W$, where H and W denote the height and width of the given input image I. This feature extractor extracts the crowd features from the image I. An enhanced ECANet (explained in subsection 3.1), captures the relevant channel features from the backbone model by avoiding the dimensionality reduction, enabling efficient channel attention. This module reduces the model's parameters and overall complexity [28]. A feature pyramid network (FPN) that fuses the multiple stage features extracted through the enhanced ECANet. The employed FPN is presented in Fig. 1. The FPN is inspired by [14], incorporating top-down approach and lateral connections. In Fig. 1, $\{P_i\}$, $i = \{1, 2, 3\}$ represent the various feature levels that the FPN structure generates where $F_{\text{pout}} = P_i \in \mathbb{R}^{\frac{R}{2^i} \times 64}$, $i = 3$ signifies the ultimate feature map produced by this FPN structure to feed the context module (CM). A context module (CM) is illustrated in Fig. 1 acting as an efficient and lightweight crowd detector, specifically a Density Map Generator (DMG), responsible for generating higher-quality density maps to facilitate density estimation. The CM is inspired by [20], known for its ability to detect small annotated heads in crowd images in a single stage for crowd counting purposes. Additionally, unlike the typical approach of processing an external multi-scale input pyramid, our employed CM is designed to be scale-invariant. This means that it can detect annotated heads at various scales within a single forward pass through the network for accurate crowd counting. Finally, the CM feature map output, $C_{\text{mout}} \in \mathbb{R}^{\frac{R}{2^i} \times 64}$, $i = 3$ followed by a simple 1×1 convolutional layer for generating a high-quality density map, $D \in \mathbb{R}^{\frac{R}{2^i} \times 1}$, $i = 3$ for accurate density estimation of the input crowd image I.

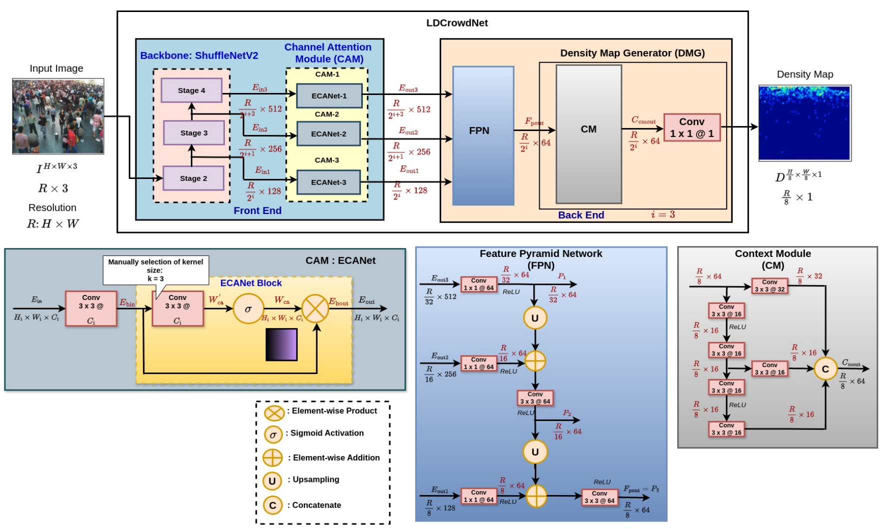

Fig. 1. LDCrowdNet's architecture includes ShuffleNetV2 [19], an enhanced ECANet for channel attention, an FPN for multi-stage feature fusion, and a context module (CM) acting as a density map generator (DMG) for generating higher-quality density maps to facilitate density estimation.

3.1 Enhanced Efficient Channel Attention Network (ECANet)

The enhanced ECANet, inspired by [28] and illustrated in Fig. 1, is a streamlined channel attention network designed to extract relevant channel features from the backbone model. Its main objective is to prevent dimensionality reduction and promote cross-channel interaction, enhancing both the effectiveness and efficiency of channel attention. Our method implements a local cross-channel interaction tactic using 2D convolution, preserving performance while substantially decreasing model complexity.

This enhanced Efficient Channel Attention (ECA) module for the proposed deep network aims to learn effective channel attention without diminishing channel dimensionality, all while efficiently capturing cross-channel interaction. It particularly emphasizes cross-channel interaction locally, where the interaction coverage (i.e., 2D convolution kernel size k) requires manual tuning. This refined coverage can be manually adjusted for convolutional blocks with varying channel numbers across different CNN architectures.

The original architecture of ECANet in [28] consists of generating channel weights from aggregated features obtained through a global average pooling (GAP). ECA accomplishes this via conducting a fast 1D convolution with size k, where k is evaluated depending on an adaptive channel dimension C mapping.

Fig. 1 offers a synopsis of the enhanced Efficient Channel Attention (ECA) module, featuring the ECANet block. Our proposed ECANet block is different from the original ECANet by removing the global average pooling (GAP) and following the aggregation of convolutional features using a preceding 2D convolutional block (instead of a fast 1D convolutional block with size k, where k is estimated according to an adaptive channel dimension C mapping) that avoids dimensionality reduction and the ECA module manually selects the kernel size k for convolutional blocks with identical channel numbers as the preceding block. Subsequently, it performs 2D convolution activated by a Sigmoid function for learning channel attention, representing the channel weights. These resulting networks are denoted as an enhanced ECANet.

$$\text{ECANet: } E_{\text{out}} = \text{ECANet}[E_{\text{in}}] \tag{1}$$

$$\text{ECANet Block: } E_{\text{bout}} = \text{ECANet Block}(E_{\text{bin}}) \tag{2}$$

$$E_{\text{out}} = E_{\text{bout}} = E_{\text{bin}} \otimes W_{\text{ca}} = E_{\text{bin}} \otimes \mathit{Sigmoid}(\mathit{Conv}(E_{\text{bin}})) \tag{3}$$

where E_{in} and E_{out} are the input and output feature maps of the ECANet and E_{bin} and E_{bout} are the input and output feature maps of the ECANet Block, respectively, and W_{ca} is the channel attention weight map, $\otimes$ is the element-wise product, *Sigmoid* is the Sigmoid activation function, and *Conv* is the convolution operation.

$$\text{FPN: } F_{\text{pout}} = \text{FPN}[E_{\text{out1}}, E_{\text{out2}}, E_{\text{out3}}] \tag{4}$$

where $F_{\text{pout}} \in \mathbb{R}^{\frac{R}{2^i} \times 64}$, $i = 3$ is the FPN output feature map, and $E_{\text{out}j}$, are the output feature maps generated by each enhanced ECANet-j, $j = \{1, 2, 3\}$ module ($E_{\text{out1}} \in \mathbb{R}^{\frac{R}{2^{i+2}} \times 512}$, $E_{\text{out2}} \in \mathbb{R}^{\frac{R}{2^{i+1}} \times 256}$, $E_{\text{out3}} \in \mathbb{R}^{\frac{R}{2^i} \times 128}$, $i = 3$), and $R : H \times W$ is the image resolution, while H and W indicate the height and width of the input image $I \in \mathbb{R}^{H \times W \times 3}$.

$$\text{CM: } C_{\text{mout}} = \text{CM}[F_{\text{pout}}] \tag{5}$$

$$\text{DMG: } D = Conv_1(C_{\text{mout}}) \tag{6}$$

where $C_{\text{mout}} \in \mathbb{R}^{\frac{R}{2^i} \times 64}$, $i = 3$ is the CM output feature map, and $Conv_1$ is the 1×1 convolution operation and $D \in \mathbb{R}^{\frac{R}{2^i} \times 1}$, $i = 3$ is the density map produced through this 1×1 convolutional layer.

3.2 Loss Function

During training, the network straightforwardly transforms from the input image into a continuous density map. We have selected the Euclidean distance metric to evaluate the disparity separating the ground truth and the generated estimated density map, following the approach used in other works [2,36,1]. The loss function is described as follows:

$$\mathcal{L}(\Theta) = \frac{1}{2N} \sum_{i=1}^{N} ||D(X_i; \Theta) - D_i^{GT}||_2^2 \tag{7}$$

where N represents the size of the training batch, and $D(X_i; \Theta)$ denotes the output density map produced by LDCrowdNet with its parameters denoted as Θ, X_i denotes the input image, although D_i^{GT} signifies the ground truth (GT) density map relevant to the input image X_i.

4 Experimental Setup

Baselines: Our proposed model, LDCrowdNet performance is evaluated versus 9 state-of-the-art models. These are LigMSANet [9], Lw-Count [18], GAPNet [4], LSANet [37], MCNN [36], SKT [17], LMSNet [29], LMSFFNet [31], and Mobile-Count [27], etc.

4.1 Implementation Details

– **Network Architecture:** We propose the architecture of LDCrowdNet, outlined in Fig. 1, comprising a front-end based on the ShuffleNetV2 backbone. Backbone ShuffleNetV2 [19] captures the input crowd image $I \in \mathbb{R}^{H \times W \times 3}$ of resolution $R : H \times W$, where H and W denote the height and width of the input image I, and extracts the crowd features through the input

image. Different stages of the backbone, ShuffleNetV2, labelled as Stage-$(i+1)$, $i = \{1, 2, 3\}$ are linked to the corresponding channel attention modules (ECANet-i, $i = \{1, 2, 3\}$) to learn the effective and efficient channel attention.
The back end incorporates a Feature Pyramid Network (FPN) to fuse the multiple-stage features extracted from the enhanced ECANet. Each input to the FPN's multi-scale layers captures feature maps from the respective ECANet-i, $i = \{1, 2, 3\}$ generating corresponding feature maps with 3 reduced channels. The highest resolution feature map is then fed to the Context Module (CM), acting as a crowd detector and Density Map Generator (DMG). Finally, a simple 1×1 convolutional layer generates a higher-quality density map for density estimation. Given that the generated output (density maps) of LDCrowdNet are reduced (1/8 of the input size), we adopt an intercubic interpolation using a scaling factor of 8 to certify the output maintains the identical resolution of the input image.

- **Training Details:** Our proposed LDCrowdNet end-to-end is trained utilizing a simple approach. During training, we utilize a ShuffleNetV2 with pretrained weights as the backbone network. The optimizer used is Adam with L2 regularization and weight decay set at 5 * 1e-4, and an initial learning rate of 1e-4. The learning rate is attenuated by 10% every 30 epochs during training. We maintain a fixed batch size of 2 for all datasets and train for 400 epochs. Evaluation of crowd counting tasks using LDCrowdNet is executed on an NVIDIA T1000 GPU as the inference platform.

- **Ground Truth Generation:** This work adopts the approach proposed in [36] and utilizes the geometry-adaptive kernels to handle the extremely crowded scenes. To create the ground truth, we apply a Gaussian kernel (normalized at 1) for blurring each head annotation, taking into account the spatial dispersion across all images in each dataset. We have generated the ground truth (GT) density map using only two popular methods: Geometry-adaptive kernels and Fixed Gaussian kernel methods: ShanghaiTech Part_A [36] and UCF_CC_50 [8] datasets density map generation through Geometry-adaptive kernels, while ShanghaiTech Part_B [36] and UCF-QNRF [5] datasets density map generation through Fixed Gaussian kernel with kernel size (standard deviation) $\sigma = 15$ and 4, respectively.

- **Data Augmentation:** To improve training robustness and prevent overfitting, we utilize random cropping of resolution 224×224 as an augmentation method. During training, LDCrowdNet receives uniformly resized inputs with resolution 224×224 for all datasets, so all input images undergo random cropping using a batch size of 2. Conversely, during testing, the uniformly resized image of resolution 224×224 is directly input utilizing a batch size of 1. To calculate the number of floating-point operations (GFLOPs) during testing, we uniformly utilize input images with VGA resolution 640×480 and a batch size of 1.

Evaluation Metrics: The Mean Absolute Error (MAE) and Mean Squared Error (MSE), since defined in [13] are utilized to evaluate the counting accuracy.

Datasets: The MAE and MSE metrics are utilized to examine three crowd counting datasets: ShanghaiTech Part_A and Part_B [36], UCF_CC_50 [8], and UCF-QNRF [5]. The statistics for these crowd counting datasets are detailed in Table 1. The chosen datasets display notable variations in object distribution, classifications, and resolution, thereby enhancing the generalization and efficacy demonstration of LDCrowdNet.

Table 1. Statistics of three crowd counting datasets.

Datasets	Images	Avg Resolution	Count Statistics			
			Total Count	Min Count	Avg Count	Max Count
ShanghaiTech Part_A [36]	482	589×868	241,677	33	501	3,139
ShanghaiTech Part_B [36]	716	768×1024	88,488	9	123	578
UCF_CC_50 [8]	50	2101×2888	63,974	49	1,279	4,633
UCF-QNRF [5]	1,535	2013×2902	1,251,642	94	815	12,865

5 Results and Discussions

This work evaluates our proposed method using three publicly available crowd counting datasets: ShanghaiTech [36], UCF_CC_50 [8], and UCF-QNRF [5]. Initially, the specifics of the implementation are presented, followed by a comprehensive evaluation and comparison against state-of-the-art methods on these three datasets. To evaluate the influence of various components within LDCrowdNet, an ablation study is conducted. This study is complemented by a comparison with the earlier state-of-the-art crowd counting networks across all three datasets. The PyTorch framework is utilized to implement the proposed model.

5.1 Quantitative Performance Analysis

Performance on ShanghaiTech Dataset: The ShanghaiTech dataset [36] includes 1,198 images totalling 330,165 annotated people (individuals), where each individual is annotated through a location near the center of their head. The dataset has been split into two sections: Part_A includes 482 images, mostly featuring densely populated scenes collected randomly from the internet, typically at 501.4 individuals per image, whereas Part_B comprises 716 images obtained from the busy metropolitan streets of Shanghai, with an average of

123.6 individuals per image, with varying scales within the same image. For SHT_A, 300 images are utilized for training and 182 for testing, while SHT_B employs 400 and 316 images for training and testing, respectively.

In this study, we examine and contrast our proposed LDCrowdNet via state-of-the-art lightweight methods in terms of counting performance, model parameters, and model computational complexity (in the context of floating-point operations), as shown in Table 2. LDCrowdNet illustrates a notable reduction in model parameters and computational complexity compared to LSANet [37] and LMSNet [29]. While our network has more parameters than LSANet [37] and LMSNet [29], it exhibits fewer GFLOPs. Additionally, LDCrowdNet's model complexity is approximately 73% of MCNN [36], a lightweight model. Despite, our method LDCrowdNet achieves better counting accuracy in the context of MAE and MSE against all other methods, therefore, for dense crowd counting, LDCrowdNet offers a superior lightweight network. The experimental illustration results from the ShanghaiTech dataset in Fig. 2 illustrate that LDCrowdNet, using its low GFLOPs model complexity, can deliver superior counting performance and generate higher-quality density maps.

Table 2. Comparison of our proposed method with the state-of-the-art methods on the ShanghaiTech Part_A/B, UCF_CC_50, and UCF-QNRF datasets.

Method	**SHT_A**		**SHT_B**		**UCF_CC_50**		**UCF-QNRF**		**Params (M)**	**GFLOPs**
	MAE	**MSE**	**MAE**	**MSE**	**MAE**	**MSE**	**MAE**	**MSE**		
LigMSANet [9]	76.6	121.4	10.9	17.5	231.5	339.7	_	_	0.63	1.3
Lw-Count [18]	69.7	100.5	10.1	12.4	239.3	307.6	149.7	238.4	0.071	2.2
GAPNet [4]	67.1	110.4	9.8	15.2	202.8	246.9	118.5	217.2	2.85	3.29
LSANet [37]	66.1	110.2	8.6	13.9	_	_	112.3	186	0.2	6.34
MCNN [36]	110.2	173.2	26.4	41.3	377.6	509.1	_	_	0.13	8.2
SKT framework: 1/4-CSRNet + SKT [17]	71.55	114.4	7.48	11.68	–	–	144.36	234.64	1.02	13.09
LMSNet [29]	62.9	108.4	8.2	13.5	223.5	281	110.7	178.7	0.73	13.7
LMSFFNet: LMSFFNet-XS [31]	85.85	139.9	9.2	15.1	105.7	120.3	112.8	201.6	4.58	14.9
MobileCount [27]	89.4	146	9	15.4	284.8	392.8	131.1	222.6	3.4	16.49
LDCrowdNet (Ours)	21.62	32.7	4.41	6.05	61.37	98.05	32.83	43.54	6.03	5.96

Performance on UCF_CC_50 Dataset: The UCF_CC_50 dataset [8] comprises 50 images featuring diverse perspectives and resolutions. Each image contains between 94 and 4543 annotated individuals, averaging 1280 persons per image. Our obtained results from the experiment are illustrated in Table 2. Noteworthy is the fact that the model complexity of LDCrowdNet is comparable to that of LSANet [37] and better to that of LMSNet [29], and MobileCount [27]. Fig. 2 demonstrates some visualization results of LDCrowdNet, indicating its counting performance.

Performance on UCF-QNRF Dataset: The UCF-QNRF dataset [5], published through the University of Central Florida in 2018, contains totalling

1535 crowd images, via 1201 images of the training set and 334 images of the test set. Specially, for training, this dataset is currently the most extensive available, and concerning the number of annotations, it is evaluating the counting networks for large-scale crowd density.

The UCF-QNRF dataset [5] provides a larger dataset suitable for dense crowd counting. The results obtained from our experiment are summarized in Table 2 demonstrate that LDCrowdNet attains fewer GFLOPs, indicating reduced model computational complexity compared to other methods. Moreover, LDCrowdNet achieved better MAE and MSE (as counting accuracy) compared to all other methods. Despite this, LDCrowdNet stands out as a better lightweight method for dense crowd counting, even in complex scenes with dense crowd density. Visualization results of LDCrowdNet are depicted in Fig. 2.

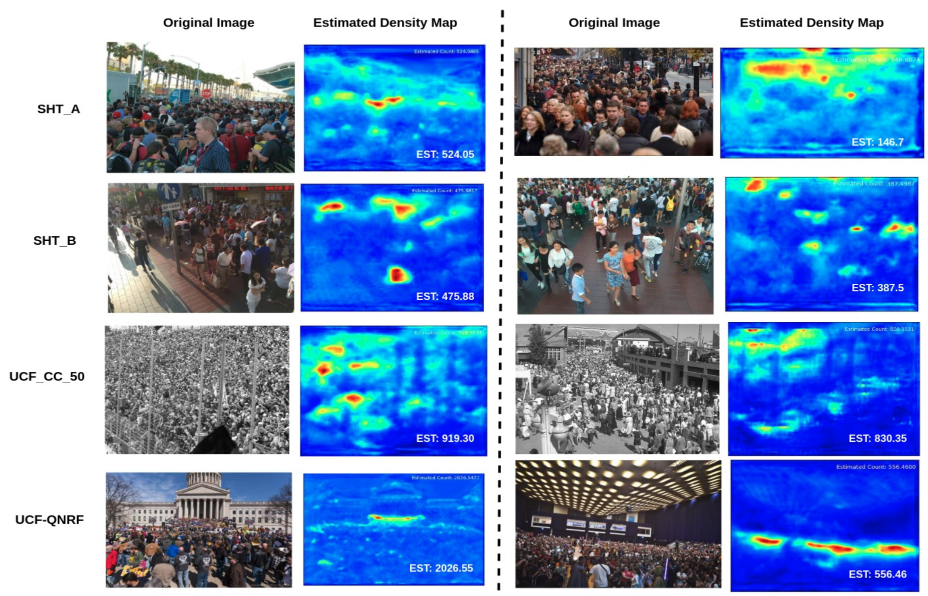

Fig. 2. Visual representations of density maps produced by our LDCrowdNet across various datasets are presented. Each row illustrates the input image, followed by the estimated (predicted) density map generated by our model. This sequence applies to images from the ShanghaiTech Part_A and Part_B, UCF_CC_50, and UCF-QNRF datasets.

Complexity Analysis: Table 2 illustrates the model's parameter count and the floating-point operations (FLOPs) in millions (M) and in giga (G), respectively, for the compared models. Our model LDCrowdNet utilizes ShuffleNetV2 [19] as the backbone, resulting in a comparatively smaller model size of 6.03 M parameters and in terms of computational efficiency, our model requires 5.96 GFLOPs. This positions our model as a lightweight crowd counter, contrasting with the other models such as LMSNet [29] (13.7 GFLOPs), MobileCount [27] (16.49 GFLOPs), LMSFFNet [31] (14.9 GFLOPs), and MCNN [36] (8.2 GFLOPs), etc.

As illustrated in Table 2, provides more detailed evidence of efficiency, indicating that our LDCrowdNet is the better lightweight model for dense crowd counting with small model size and reduced computational model complexity, requiring less computation.

5.2 Ablation Study

To validate the efficacy of our proposed method, we performed the ablation experimentations on the ShanghaiTech dataset. The input consists of uniformly resized images of resolution 224×224 for testing the GFLOPs with a batch size of 1.

Effectiveness of the Backbones: To evaluate the efficacy of the various models on counting accuracy and model complexity, we compared four different versions of ShuffleNetV2 [19] lightweight backbones. The results from the ablation experiments in Table 3 demonstrate that ShuffleNetV2_x1.0 attains the least Mean Absolute Error (MAE) and Mean Squared Error (MSE) on both ShanghaiTech Part_A (SHT_A) and Part_B (SHT_B), despite having slightly more parameters and GFLOPs compared to ShuffleNetV2_x0.5. While ShuffleNetV2_x2.0 has higher parameters and GFLOPs than ShuffleNetV2_x1.0, its overall performance is inferior. Considering the practical applications of the model in dense crowd counting, ShuffleNetV2_x1.0 is selected as the backbone for subsequent experiments.

Table 3. Ablation on the backbones

Backbone	SHT_A		SHT_B		Params (M)	GFLOPs
	MAE	MSE	MAE	MSE		
ShuffleNetV2_x0.5 [19]	23.9	35.2	10.03	12	1.15	1.85
ShuffleNetV2_x1.0 [19]	21.62	32.7	4.41	6.05	6.03	5.96
ShuffleNetV2_x1.5 [19]	22.9	33.37	6.08	7.62	13.66	12.32
ShuffleNetV2_x2.0 [19]	25.24	37.04	6.82	8.38	26.07	22.58

Effectiveness of the Channel Attention Module: ECANet: To investigate the effectiveness of different attention mechanisms, the impact on LDCrowdNet's counting accuracy and model complexity, we performed experiments with and without the channel attention module (i.e., enhanced ECANet), while keeping other network structures constant. The results on the ablation experiments in Table 4 demonstrate that LDCrowdNet without enhanced ECANet attains the best Mean Absolute Error (MAE) and Mean Squared Error (MSE) on ShanghaiTech Part_A (SHT_A), with lower parameters and GFLOPs compared to when using enhanced ECANet. However, it exhibits higher MAE and MSE on ShanghaiTech Part_B (SHT_B) compared to when enhanced ECANet is included. This suggests that ECANet effectively lowers the number of parameters and computational complexity while enhancing connectivity between feature map channels. It allocates each channel more precise weights, resulting in density maps that more closely resemble the ground truth (GT) density map.

Table 4. Ablation on the channel attention module: ECANet

Channel Attention Module: ECANet	**SHT_A**		**SHT_B**		**Params (M)**	**GFLOPs**
	MAE	**MSE**	**MAE**	**MSE**		
w Enhanced ECANet	21.62	32.7	4.41	6.05	6.03	5.96
w/o Enhanced ECANet	21.3	32.54	6.07	7.13	0.94	2.48

Effectiveness of the CM: To validate the effectiveness of the context module (CM), we conducted experiments both with and without CM. The results on the ablation experiments in Table 5 demonstrate that without CM, our model attains higher Mean Absolute Error (MAE) and Mean Squared Error (MSE) on both ShanghaiTech Part_A and Part_B (SHT_A and SHT_B), with lower parameters and GFLOPs compared to when CM is included. This underscores CM's role in significantly reducing the model complexity while serving as a crowd detector and functioning as a Density Map Generator (DMG), crucial for generating higher-quality density maps for density estimation.

Table 5. Ablation experiment on the context module (CM)

Context Module (CM)	**SHT_A**		**SHT_B**		**Params (M)**	**GFLOPs**
	MAE	**MSE**	**MAE**	**MSE**		
w CM	21.62	32.7	4.41	6.05	6.03	5.96
w/o CM	22.13	33.05	7.3	9.35	6	5.64

6 Conclusion

In the dense crowd counting application, we employed a ShuffleNetV2 as the backbone for our simplified lightweight network. We propose a lightweight model

LDCrowdNet, specifically designed for dense crowd counting and density estimation. We proposed an enhanced ECANet for efficient channel attention, and we employed a feature pyramid network (FPN), and a context module (CM) that serves as a crowd detector, i.e., acting as a Density Map Generator (DMG). Extensive experiments were conducted across three publicly available crowd counting benchmark datasets to validate the effectiveness of our LDCrowdNet. The model is suitable for performing dense crowd counting tasks due to its reduced parameters and model complexity (GFLOPs) without sacrificing accuracy. Future work aims to explore further methods for model complexity reduction and expand applications for developing advanced lightweight networks of deep learning for crowd counting and real-time object counting tasks.

References

1. Babu Sam, D., Surya, S., Venkatesh Babu, R.: Switching convolutional neural network for crowd counting. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 5744–5752 (2017)
2. Boominathan, L., Kruthiventi, S.S., Babu, R.V.: Crowdnet: A deep convolutional network for dense crowd counting. In: Proceedings of the 24th ACM international conference on Multimedia. pp. 640–644 (2016)
3. Gao, G., Gao, J., Liu, Q., Wang, Q., Wang, Y.: Cnn-based density estimation and crowd counting: A survey. arXiv preprint arXiv:2003.12783 (2020)
4. Guo, X., Song, K., Gao, M., Zhai, W., Li, Q., Jeon, G.: Crowd counting in smart city via lightweight ghost attention pyramid network. Future Generation Computer Systems **147**, 328–338 (2023)
5. H. Idrees, M. Tayyab, K.A.D.Z.S.A.M.N.R.M.S.: Composition loss for counting, density map estimation and localization in dense crowds. In: Proceedings of the European Conference on Computer Vision (ECCV) (2018)
6. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 770–778 (2016)
7. Howard, A.G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., Adam, H.: Mobilenets: Efficient convolutional neural networks for mobile vision applications. arXiv preprint arXiv:1704.04861 (2017)
8. Idrees, H., Saleemi, I., Seibert, C., Shah, M.: Multi-source multi-scale counting in extremely dense crowd images. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 2547–2554 (2013)
9. Jiang, G., Wu, R., Huo, Z., Zhao, C., Luo, J.: Ligmsanet: Lightweight multi-scale adaptive convolutional neural network for dense crowd counting. Expert Systems with Applications **197**, 116662 (2022)
10. Jingying, W.: A survey on crowd counting methods and datasets. In: Advances in Computer, Communication and Computational Sciences: Proceedings of IC4S 2019. pp. 851–863. Springer (2021)
11. Kang, D., Ma, Z., Chan, A.B.: Beyond counting: comparisons of density maps for crowd analysis tasks—counting, detection, and tracking. IEEE Transactions on Circuits and Systems for Video Technology **29**(5), 1408–1422 (2018)
12. Li, T., Chang, H., Wang, M., Ni, B., Hong, R., Yan, S.: Crowded scene analysis: A survey. IEEE transactions on circuits and systems for video technology **25**(3), 367–386 (2014)

13. Li, Y., Zhang, X., Chen, D.: Csrnet: Dilated convolutional neural networks for understanding the highly congested scenes. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 1091–1100 (2018)
14. Lin, T.Y., Dollár, P., Girshick, R., He, K., Hariharan, B., Belongie, S.: Feature pyramid networks for object detection. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 2117–2125 (2017)
15. Liu, J., Jin, H., Xu, G., Lin, M., Wu, T., Nour, M., Alenezi, F., Alhudhaif, A., Polat, K.: Aliasing black box adversarial attack with joint self-attention distribution and confidence probability. Expert Systems with Applications **214**, 119110 (2023)
16. Liu, J., Zheng, S., Xu, G., Lin, M.: Cross-domain sentiment aware word embeddings for review sentiment analysis. International Journal of Machine Learning and Cybernetics **12**, 343–354 (2021)
17. Liu, L., Chen, J., Wu, H., Chen, T., Li, G., Lin, L.: Efficient crowd counting via structured knowledge transfer. In: Proceedings of the 28th ACM international conference on multimedia. pp. 2645–2654 (2020)
18. Liu, Y., Cao, G., Shi, H., Hu, Y.: Lw-count: An effective lightweight encoding-decoding crowd counting network. IEEE Transactions on Circuits and Systems for Video Technology **32**(10), 6821–6834 (2022)
19. Ma, N., Zhang, X., Zheng, H.T., Sun, J.: Shufflenet v2: Practical guidelines for efficient cnn architecture design. In: Proceedings of the European conference on computer vision (ECCV). pp. 116–131 (2018)
20. Najibi, M., Samangouei, P., Chellappa, R., Davis, L.S.: Ssh: Single stage headless face detector. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV) (Oct 2017)
21. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.C.: Mobilenetv2: Inverted residuals and linear bottlenecks. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 4510–4520 (2018)
22. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556 (2014)
23. Sindagi, V.A., Patel, V.M.: Generating high-quality crowd density maps using contextual pyramid cnns. In: Proceedings of the IEEE international conference on computer vision. pp. 1861–1870 (2017)
24. Sindagi, V.A., Patel, V.M.: A survey of recent advances in cnn-based single image crowd counting and density estimation. Pattern Recognition Letters **107**, 3–16 (2018)
25. Sun, K., Xiao, B., Liu, D., Wang, J.: Deep high-resolution representation learning for human pose estimation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 5693–5703 (2019)
26. Topkaya, I.S., Erdogan, H., Porikli, F.: Counting people by clustering person detector outputs. In: 2014 11th IEEE international conference on advanced video and signal-based surveillance (AVSS). pp. 313–318. IEEE (2014)
27. Wang, P., Gao, C., Wang, Y., Li, H., Gao, Y.: Mobilecount: An efficient encoder-decoder framework for real-time crowd counting. Neurocomputing **407**, 292–299 (2020)
28. Wang, Q., Wu, B., Zhu, P., Li, P., Zuo, W., Hu, Q.: Eca-net: Efficient channel attention for deep convolutional neural networks. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 11534–11542 (2020)
29. Xi, M., Yan, H.: Lightweight multi-scale network with attention for accurate and efficient crowd counting. The Visual Computer pp. 1–14 (2023)

30. Xu, H., Cai, Z., Li, R., Li, W.: Efficient citycam-to-edge cooperative learning for vehicle counting in its. IEEE Transactions on Intelligent Transportation Systems **23**(9), 16600–16611 (2022)
31. Yi, J., Shen, Z., Chen, F., Zhao, Y., Xiao, S., Zhou, W.: A lightweight multiscale feature fusion network for remote sensing object counting. IEEE Transactions on Geoscience and Remote Sensing **61**, 1–13 (2023)
32. Zhan, B., Monekosso, D.N., Remagnino, P., Velastin, S.A., Xu, L.Q.: Crowd analysis: a survey. Machine Vision and Applications **19**, 345–357 (2008)
33. Zhang, C., Kang, K., Li, H., Wang, X., Xie, R., Yang, X.: Data-driven crowd understanding: A baseline for a large-scale crowd dataset. IEEE Transactions on Multimedia **18**(6), 1048–1061 (2016)
34. Zhang, C., Li, H., Wang, X., Yang, X.: Cross-scene crowd counting via deep convolutional neural networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 833–841 (2015)
35. Zhang, S., Wu, G., Costeira, J.P., Moura, J.M.: Understanding traffic density from large-scale web camera data. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 5898–5907 (2017)
36. Zhang, Y., Zhou, D., Chen, S., Gao, S., Ma, Y.: Single-image crowd counting via multi-column convolutional neural network. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 589–597 (2016)
37. Zhu, F., Yan, H., Chen, X., Li, T.: Real-time crowd counting via lightweight scale-aware network. Neurocomputing **472**, 54–67 (2022)